

```
#include <stdio.h>
```

```
int main (int argc, char* argv[]) {  
    printf("Hello World \n");  
    return 0;  
}
```

Annotations for the code above:

- `int`: Rückgabe d. Wertes
- `main`: Funktionsname
- `(int argc, char* argv[])`: Formalparameterliste
- `int` (under `argc`): DT (Datentyp)
- `char*` (under `argv`): name
- `{`: Block
- `printf("Hello World \n");`: Funktionsaufruf
- `"Hello World \n"`: Aktualparameterliste

statisch $\leftrightarrow$ dynamisch
zur Compilezeit $\leftrightarrow$ zur Laufzeit

Diagram illustrating the mapping of command-line arguments to `argc` and `argv`:

Command: `C:\> ping 10 -n 10 100 10`

Mapping:

- `argc = 4` (points to the first four arguments)
- `argv` (points to the array of argument strings)
- Arguments: `10`, `microsoft.com`, `-n 10`, `100 10`

```
for (int i = 0; i < argc; i = i + 1) {  
    printf("%d arg: %s \n", i, argv[i]);  
}
```

Output:

```
=> 0 arg: C:\>ping.exe  
1 arg: microsoft.com  
2 arg: -n
```

3 arg: 100

if (x ≠ 5) if (5 ≠ x)

 L-Value r-Value
 ↓ ↙
 ↘
 L-Value
 ↘
 Fehler

Programm

Celsius To Kelvin